CollegeBoard

AP® Macroeconomics Practice Exam

From the 2016 Administration

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Note: This publication shows the page numbers that appeared in the *2015–16 AP Exam Instructions* book and in the actual exam. This publication was not repaginated to begin with page 1.

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Exam Instructions

The following contains instructions taken from the *2015–16 AP Exam Instructions* book.

AP® Macroeconomics Exam

Regularly Scheduled Exam Date: Wednesday afternoon, May 11, 2016

Late-Testing Exam Date: Wednesday morning, May 18, 2016

Section I Total Time: 1 hr. 10 min. Section II Total Time: 1 hr.

AP® Microeconomics Exam

Regularly Scheduled Exam Date: Friday morning, May 13, 2016

Late-Testing Exam Date: Wednesday afternoon, May 18, 2016

Section I Total Time: 1 hr. 10 min. Section II Total Time: 1 hr.

Section I

Total Time: 1 hour 10 minutes
Calculator not permitted
Number of Questions: 60*
Percent of Total Score: 66.67%
Writing Instrument: Pencil required

**The number of questions may vary slightly depending on the form of the exam.*

Section II

Total Time: 1 hour
Calculator not permitted
Number of Questions: 3 required free-response questions
(10-minute reading period, 50-minute writing period)
Percent of Total Score: 33.33%
Writing Instrument: Pen with black or dark blue ink

What Proctors Need to Bring to This Exam

- Exam packets
- Answer sheets
- AP Student Packs
- *2015-16 AP Coordinator's Manual*
- This book — *AP Exam Instructions*
- AP Exam Seating Chart template(s)
- School Code and Home-School/Self-Study Codes
- Pencil sharpener
- Container for students' electronic devices (if needed)
- Extra No. 2 pencils with erasers
- Extra pens with black or dark blue ink
- Lined paper
- Stapler
- Watch
- Signs for the door to the testing room
  - "Exam in Progress"
  - "Cell phones are prohibited in the testing room"

SECTION I: Multiple Choice

Do not begin the exam instructions below until you have completed the appropriate General Instructions for your group.

Make sure you begin the exam at the designated time. Remember, you must complete a seating chart for this exam. See pages 305–306 for a seating chart template and instructions. See the *2015-16 AP Coordinator's Manual* for exam seating requirements (pages 49–52).

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Macroeconomics

If you are giving the regularly scheduled exam, say:

It is Wednesday afternoon, May 11, and you will be taking the AP Macroeconomics Exam.

If you are giving the alternate exam for late testing, say:

It is Wednesday morning, May 18, and you will be taking the AP Macroeconomics Exam.

Microeconomics

If you are giving the regularly scheduled exam, say:

It is Friday morning, May 13, and you will be taking the AP Microeconomics Exam.

If you are giving the alternate exam for late testing, say:

It is Wednesday afternoon, May 18, and you will be taking the AP Microeconomics Exam.

In a moment, you will open the packet that contains your exam materials. By opening this packet, you agree to all of the AP Program's policies and procedures outlined in the *2015-16 Bulletin for AP Students and Parents*. You may now remove the shrinkwrap from your exam packet and take out the Section I booklet, but do not open the booklet or the shrinkwrapped Section II materials. Put the white seals aside. . . .

Carefully remove the AP Exam label found near the top left of your exam booklet cover. Now place it on page 1 of your answer sheet on the light blue box near the top right-hand corner that reads "AP Exam Label."

If students accidentally place the exam label in the space for the number label or vice versa, advise them to leave the labels in place. They should not try to remove the label; their exam can still be processed correctly.

Read the statements on the front cover of Section I and look up when you have finished. . . .

Sign your name and write today's date. Look up when you have finished. . . .

Now print your full legal name where indicated. Are there any questions? . . .

Turn to the back cover of your exam booklet and read it completely. Look up when you have finished. . . .

Are there any questions? . . .

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You will now take the multiple-choice portion of the exam. You should have in front of you the multiple-choice booklet and your answer sheet. Open your answer sheet to page 2. You may never discuss these specific multiple-choice questions at any time in any form with anyone, including your teacher and other students. If you disclose these questions through any means, your AP Exam score will be canceled.

You must complete the answer sheet using a No. 2 pencil only. Mark all of your responses beginning on page 2 of your answer sheet, one response per question. Completely fill in the circles. If you need to erase, do so carefully and completely. No credit will be given for anything written in the exam booklet. Scratch paper is not allowed, but you may use the margins or any blank space in the exam booklet for scratch work. Calculators are not allowed on any part of this exam. Are there any questions? . . .

You have 1 hour and 10 minutes for this section. Open your Section I booklet and begin.

Note Start Time here _____. Note Stop Time here _____. Check that students are marking their answers in pencil on their answer sheets and that they are not looking at their shrinkwrapped Section II booklets. After 1 hour, say:

There are 10 minutes remaining.

After 10 minutes, say:

Stop working. Close your booklet and put your answer sheet on your desk, face up. Make sure you have your AP number label and an AP Exam label on page 1 of your answer sheet. Sit quietly while I collect your answer sheets.

Collect an answer sheet from each student. Check that each answer sheet has an AP number label and an AP Exam label. After all answer sheets have been collected, say:

Now you must seal your exam booklet using the white seals you set aside earlier. Remove the white seals from the backing and press one on each area of your exam booklet cover marked "PLACE SEAL HERE." Fold each seal over the back cover. When you have finished, place the booklet on your desk, face up. I will now collect your Section I booklet. . . .

Collect a Section I booklet from each student. Check that each student has signed the front cover of the sealed Section I booklet.

There is a 10-minute break between Sections I and II. When all Section I materials have been collected and accounted for and you are ready for the break, say:

Please listen carefully to these instructions before we take a 10-minute break. All items you placed under your chair at the beginning of this exam must stay there, and you are not permitted to open or access them in any way. Leave your shrinkwrapped Section II packet on your desk during the break. You are not allowed to consult teachers, other students, notes, or textbooks during the break. You may not make phone calls, send text messages, check email, use a social networking site, or access any electronic

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or communication device. Remember, you may never discuss the multiple-choice questions at any time in any form with anyone, including your teacher and other students. If you disclose these questions through any means, your AP Exam score will be canceled. Are there any questions? . . .

You may begin your break. Testing will resume at _____.

SECTION II: Free Response

After the break, say:

May I have everyone's attention? Place your Student Pack on your desk. . . .

You may now remove the shrinkwrap from the Section II packet, but do not open the exam booklet until you are told to do so. . . .

Read the bulleted statements on the front cover of the exam booklet. Look up when you have finished. . . .

Now take an AP number label from your Student Pack and place it on the shaded box. If you don't have any AP number labels, write your AP number in the box. Look up when you have finished. . . .

Read the last statement. . . .

Using a pen with black or dark blue ink, print the first, middle, and last initials of your legal name in the boxes and print today's date where indicated. This constitutes your signature and your agreement to the statements on the front cover. . . .

Turn to the back cover and, using your pen, complete Item 1 under "Important Identification Information." Print the first two letters of your <u>last</u> name and the first letter of your <u>first</u> name in the boxes. Look up when you have finished. . . .

In Item 2, print your date of birth in the boxes. . . .

In Item 3, write the school code you printed on the front of your Student Pack in the boxes. . . .

Read Item 4. . . .

Are there any questions? . . .

I need to collect the Student Pack from anyone who will be taking another AP Exam. You may keep it only if you are not taking any other AP Exams this year. If you have no other AP Exams to take, place your Student Pack under your chair now. . . .

Read the information on the back cover of the exam booklet. Do not open the booklet until you are told to do so. Look up when you have finished. . . .

Collect the Student Packs. Then say:

Are there any questions? . . .

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The total Section II time is 1 hour. This includes a 10-minute reading period. The reading period is designed to provide you with time to develop thoughtful, well-organized responses. You are advised to take advantage of the reading period to plan what you will write. You may begin writing your exam responses before the reading period is over. You may make notes on page 3 and the pages that contain the exam questions, but your responses must be written on the designated lined pages. Are there any questions? . . .

You are responsible for pacing yourself and may proceed freely from one question to the next. You must write your answers in the Section II booklet using a pen with black or dark blue ink. You are not permitted to use other colored pens or pencils to draw graphs or diagrams. If you need more paper to complete your responses, raise your hand. At the top of each extra sheet of paper you use, be sure to write only:

- **your AP number, and**
- **the question number you are working on.**

Make sure to begin your response on the lined page directly following the first appearance of the question and continue your response on the additional lined pages that follow. Remember to write your answers on the lined pages provided for each question; only responses written on the lined pages will be scored.

You may now open the Section II booklet and begin the 10-minute reading period.

Note Start Time here _____. Note Stop Time here _____. After 10 minutes, say:

The reading period is over. You have 50 minutes remaining to complete Section II.

Note Start Time here _____. Note Stop Time here _____. Check that students are using pens and that they are writing their answers in their exam booklets. After 40 minutes, say:

There are 10 minutes remaining.

After 10 minutes, say:

Stop working and close your exam booklet. Place it on your desk, face up. . . .

If any students used extra paper for a question in the free-response section, have those students staple the extra sheet(s) to the first page corresponding to that question in their exam booklets. Complete an Incident Report. A single Incident Report may be completed for multiple students per exam subject per administration (regular or late testing) as long as all of the required information is provided. Include all exam booklets with extra sheets of paper in an Incident Report return envelope (see page 60 of the *2015-16 AP Coordinator's Manual* for complete details). Then say:

Remain in your seat, without talking, while the exam materials are collected. . . .

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Collect a Section II booklet from each student. Check for the following:

- Exam booklet front cover: The student placed an AP number label on the shaded box and printed his or her initials and today's date.
- Exam booklet back cover: The student completed the "Important Identification Information" area.

When all exam materials have been collected and accounted for, return to students any electronic devices you may have collected before the start of the exam.

If you are giving the regularly scheduled exam, say:

You may not discuss or share these specific free-response questions with anyone unless they are released on the College Board website in about two days. Your AP Exam score results will be available online in July.

If you are giving the alternate exam for late testing, say:

None of the questions in this exam may ever be discussed or shared in any way at any time. Your AP Exam score results will be available online in July.

If any students completed the AP number card at the beginning of this exam, say:

Please remember to take your AP number card with you. You will need the information on this card to view your scores and order AP score reporting services online.

Then say:

You are now dismissed.

All exam materials must be placed in secure storage until they are returned to the AP Program after your school's last administration. Before storing materials, check the "School Use Only" section on page 1 of the answer sheet and:

- Fill in the appropriate section number circle in order to access a separate AP Instructional Planning Report (for regularly scheduled exams only) or subject score roster at the class section or teacher level. See "Post-Exam Activities" in the *2015-16 AP Coordinator's Manual*.
- Check your list of students who are eligible for fee reductions and fill in the appropriate circle on their registration answer sheets.

Be sure to give the completed seating chart to the AP Coordinator. Schools must retain seating charts for at least six months (unless the state or district requires that they be retained for a longer period of time). Schools should not return any seating charts in their exam shipments unless they are required as part of an Incident Report.

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Student Answer Sheet for the Multiple-Choice Section

Use this section to capture student responses. (Note that the following answer sheet is a sample, and may differ from one used in an actual exam.)

AP Answer Sheet 2016

AP Number Label (from Student Pack)

B123456789T

AP Exam Label (from Section I Booklet)

CollegeBoard

COMPLETE THIS AREA AT EVERY EXAM. USE NO. 2 PENCIL ONLY

To maintain the security of the exam and the validity of my AP score, I will allow no one else to see the multiple-choice questions. I will seal the multiple-choice booklet when asked to do so, and I will not discuss these questions with anyone at any time after completing the section. I am aware of and agree to the AP Program's policies and procedures as outlined in the *2015-16 Bulletin for AP Students and Parents*, including using testing accommodations (e.g., extended time, computer, etc.) only if I have been preapproved by College Board Services for Students with Disabilities.

A. SIGNATURE Sign your legal name as it will appear on your college applications. Date

B. LEGAL NAME Omit apostrophes, Jr., II.

Legal Last Name — First 15 Letters | Legal First Name — First 12 Letters | MI

C. YOUR AP NUMBER

D. EXAM DATE Month Day

E. EXAM START TIME AM PM

F. MULTIPLE-CHOICE BOOKLET SERIAL NUMBER S

G. ONLINE PROVIDER CODE

COMPLETE THIS AREA ONLY ONCE.

I. AREA CODE AND PHONE NUMBER

INTERNATIONAL PHONE

J. SCHOOL YOU ATTEND

SCHOOL CODE

School Name

City

State

Country

K. DATE OF BIRTH

Month: Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec

Day Year

L. SOCIAL SECURITY NUMBER (Optional)

M. COLLEGE TO RECEIVE YOUR AP SCORE REPORT

COLLEGE CODE

Using the college code listed in the AP Student Pack, indicate the **ONE** college that you want to receive your AP score report.

College Name

City

State

Country

N. CURRENT GRADE LEVEL

Not yet in 9th grade
9th
10th
11th
12th
No longer in high school

O. STUDENT SEARCH SERVICE®

Colleges and scholarship programs may request your information to inform you of educational opportunities and financial aid.

Would you like us to supply your information?

Yes No

If you don't answer and previously chose to participate in this service, we will continue providing your information.

H. AP EXAM I AM TAKING USING THIS ANSWER SHEET

Exam Name: Form: Form Code:

785208

SCHOOL USE ONLY

Section Number: 1 2 3 4 5 6 7 8 9

Fee Reduction Granted: 1 Option 1 2 Option 2

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COMPLETE THIS AREA AT EACH EXAM (IF APPLICABLE).

P. SURVEY QUESTIONS — Answer the survey questions in the AP Student Pack. Do not put responses to exam questions in this section.

1	A	B	C	D	E	F	G	H	I	4	A	B	C	D	E	F	G	H	I	7	A	B	C	D	E	F	G	H	I
2	A	B	C	D	E	F	G	H	I	5	A	B	C	D	E	F	G	H	I	8	A	B	C	D	E	F	G	H	I
3	A	B	C	D	E	F	G	H	I	6	A	B	C	D	E	F	G	H	I	9	A	B	C	D	E	F	G	H	I

Q. LANGUAGE — Do not complete this section unless instructed to do so.

If this answer sheet is for the French Language and Culture, German Language and Culture, Italian Language and Culture, Spanish Language and Culture, or Spanish Literature and Culture Exam, please answer the following questions. Your responses will not affect your score.

1. Have you lived or studied for one month or more in a country where the language of the exam you are now taking is spoken?

◯ Yes ◯ No

2. Do you regularly speak or hear the language at home?

◯ Yes ◯ No

QUESTIONS 1–75

Indicate your answers to the exam questions in this section (pages 2 and 3). Mark only one response per question for Questions 1 through 120. If a question has only four answer options, do not mark option E. Answers written in the multiple-choice booklet will not be scored.

COMPLETE MARK ●

EXAMPLES OF INCOMPLETE MARKS

You must use a No. 2 pencil and marks must be complete. Do not use a mechanical pencil. It is very important that you fill in the entire circle darkly and completely. If you change your response, erase as completely as possible. Incomplete marks or erasures may affect your score.

1	A	B	C	D	E	26	A	B	C	D	E	51	A	B	C	D	E
2	A	B	C	D	E	27	A	B	C	D	E	52	A	B	C	D	E
3	A	B	C	D	E	28	A	B	C	D	E	53	A	B	C	D	E
4	A	B	C	D	E	29	A	B	C	D	E	54	A	B	C	D	E
5	A	B	C	D	E	30	A	B	C	D	E	55	A	B	C	D	E
6	A	B	C	D	E	31	A	B	C	D	E	56	A	B	C	D	E
7	A	B	C	D	E	32	A	B	C	D	E	57	A	B	C	D	E
8	A	B	C	D	E	33	A	B	C	D	E	58	A	B	C	D	E
9	A	B	C	D	E	34	A	B	C	D	E	59	A	B	C	D	E
10	A	B	C	D	E	35	A	B	C	D	E	60	A	B	C	D	E
11	A	B	C	D	E	36	A	B	C	D	E	61	A	B	C	D	E
12	A	B	C	D	E	37	A	B	C	D	E	62	A	B	C	D	E
13	A	B	C	D	E	38	A	B	C	D	E	63	A	B	C	D	E
14	A	B	C	D	E	39	A	B	C	D	E	64	A	B	C	D	E
15	A	B	C	D	E	40	A	B	C	D	E	65	A	B	C	D	E
16	A	B	C	D	E	41	A	B	C	D	E	66	A	B	C	D	E
17	A	B	C	D	E	42	A	B	C	D	E	67	A	B	C	D	E
18	A	B	C	D	E	43	A	B	C	D	E	68	A	B	C	D	E
19	A	B	C	D	E	44	A	B	C	D	E	69	A	B	C	D	E
20	A	B	C	D	E	45	A	B	C	D	E	70	A	B	C	D	E
21	A	B	C	D	E	46	A	B	C	D	E	71	A	B	C	D	E
22	A	B	C	D	E	47	A	B	C	D	E	72	A	B	C	D	E
23	A	B	C	D	E	48	A	B	C	D	E	73	A	B	C	D	E
24	A	B	C	D	E	49	A	B	C	D	E	74	A	B	C	D	E
25	A	B	C	D	E	50	A	B	C	D	E	75	A	B	C	D	E

ETS USE ONLY

Exam		0	1	2	3	4	5	6	7	8	9
		0	1	2	3	4	5	6	7	8	9
Exam		0	1	2	3	4	5	6	7	8	9
		0	1	2	3	4	5	6	7	8	9

SELECTED MEDIA EXAMS	R	W	O	OTHER EXAMS	R	W	O
PT02				TOTAL			
PT03				Subscore (if applicable)			
PT04				Subscore (if applicable)			

DO NOT WRITE IN THIS AREA

QUESTIONS 76–120

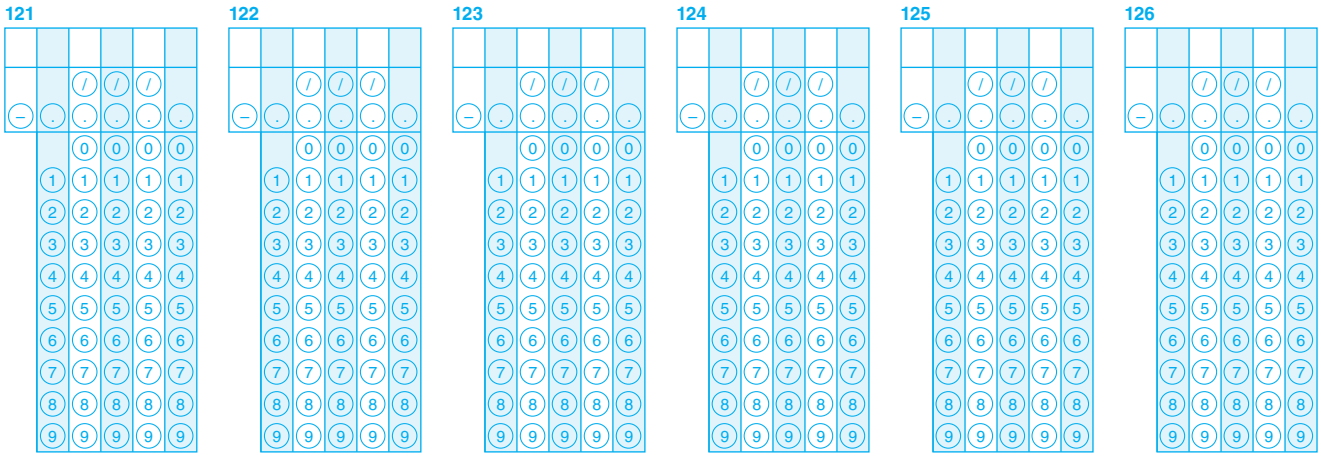
Be sure each mark is dark and completely fills the circle. If a question has only four answer options, do not mark option E.

76	A	B	C	D	E	91	A	B	C	D	E	106	A	B	C	D	E
77	A	B	C	D	E	92	A	B	C	D	E	107	A	B	C	D	E
78	A	B	C	D	E	93	A	B	C	D	E	108	A	B	C	D	E
79	A	B	C	D	E	94	A	B	C	D	E	109	A	B	C	D	E
80	A	B	C	D	E	95	A	B	C	D	E	110	A	B	C	D	E
81	A	B	C	D	E	96	A	B	C	D	E	111	A	B	C	D	E
82	A	B	C	D	E	97	A	B	C	D	E	112	A	B	C	D	E
83	A	B	C	D	E	98	A	B	C	D	E	113	A	B	C	D	E
84	A	B	C	D	E	99	A	B	C	D	E	114	A	B	C	D	E
85	A	B	C	D	E	100	A	B	C	D	E	115	A	B	C	D	E
86	A	B	C	D	E	101	A	B	C	D	E	116	A	B	C	D	E
87	A	B	C	D	E	102	A	B	C	D	E	117	A	B	C	D	E
88	A	B	C	D	E	103	A	B	C	D	E	118	A	B	C	D	E
89	A	B	C	D	E	104	A	B	C	D	E	119	A	B	C	D	E
90	A	B	C	D	E	105	A	B	C	D	E	120	A	B	C	D	E

QUESTIONS 121–126

For Students Taking AP Biology

Write your answer in the boxes at the top of the griddable area and fill in the corresponding circles.
Mark only one circle in any column. You will receive credit only if the circles are filled in correctly.



QUESTIONS 131–142

For Students Taking AP Physics 1 or AP Physics 2

Mark two responses per question. You will receive credit only if both correct responses are selected.

131	A	B	C	D	135	A	B	C	D	139	A	B	C	D
132	A	B	C	D	136	A	B	C	D	140	A	B	C	D
133	A	B	C	D	137	A	B	C	D	141	A	B	C	D
134	A	B	C	D	138	A	B	C	D	142	A	B	C	D

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DO NOT WRITE IN THIS AREA

COMPLETE THIS AREA ONLY ONCE.

R. YOUR MAILING ADDRESS

Use the address abbreviations from your AP Student Pack. Fill in only one circle per column. Indicate a space in your address by leaving a blank box; do not grid that column.

STREET ADDRESS (include street number, street name, apartment number, etc.)

CITY

ZIP OR POSTAL CODE

COUNTRY CODE

STATE

AK	HI	MI	NY	VT
AL	IA	MN	OH	WA
AR	ID	MO	OK	WI
AZ	IL	MS	OR	WV
CA	IN	MT	PA	WY
CO	KS	NC	RI	Puerto Rico
CT	KY	ND	SC	
DC	LA	NE	SD	AA
DE	MA	NH	TN	AE
FL	MD	NJ	TX	AP
GA	ME	NM	UT	Other
		NV	VA	

V. SEX

- Female
- Male

W. WHICH LANGUAGE DO YOU KNOW BEST?

- English
- English and another language about the same
- Another language

X. RACIAL/ETHNIC GROUP

Please answer both questions about Hispanic origin and about race. For the following questions about your identity, Hispanic origins are not races.

(You may mark all that apply.)

a. Are you of Hispanic, Latino, or Spanish origin?

- No, not of Hispanic, Latino, or Spanish origin
- Yes, Cuban
- Yes, Mexican
- Yes, Puerto Rican
- Yes, another Hispanic, Latino, or Spanish origin

b. What is your race?

- American Indian or Alaska Native
- Asian (including Indian subcontinent and Philippines origin)
- Black or African American (including Africa and Afro-Caribbean origin)
- Native Hawaiian or other Pacific Islander
- White (including Middle Eastern origin)

Y. PARENTAL EDUCATION LEVEL

In the first column, indicate the highest level of education of one parent/guardian, and indicate whether this is your mother/female guardian or father/male guardian. Then, if applicable, indicate the highest level of education of your other parent/guardian in the second column, and indicate whether this is your mother/female guardian or father/male guardian.

- **Mother or female guardian**
- **Father or male guardian**

- Grade school
- Some high school
- High school diploma or equivalent
- Vocational or trade school
- Some college
- Associate or two-year degree
- Bachelor's or four-year degree
- Some graduate or professional school
- Graduate or professional degree

S. FOR STUDENTS OUTSIDE THE UNITED STATES ONLY

If your address does not fit in the spaces provided in Item R, fill in as many circles as you can, then fill in the circle in Item S and print the remainder of your address in the spaces provided.

Address	City	State or Province	Country	ZIP or Postal Code

T. STUDENT IDENTIFIER (Student ID Number)

U. EMAIL ADDRESS

By providing your email address, you are granting the College Board permission to use your email in accordance with the policies in the *2015-16 Bulletin for AP Students and Parents*.

Section I: Multiple-Choice Questions

This is the multiple-choice section of the 2016 AP exam.
It includes cover material and other administrative instructions
to help familiarize students with the mechanics of the exam.
(Note that future exams may differ in look from the following content.)

AP® Macroeconomics Exam

SECTION I: Multiple Choice

2016

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

At a Glance

Total Time
1 hour, 10 minutes
Number of Questions
60
Percent of Total Score
66.67%
Writing Instrument
Pencil required
Electronic Device
None allowed

Instructions

Section I of this exam contains 60 multiple-choice questions. Fill in only the circles for numbers 1 through 60 on your answer sheet.

Indicate all of your answers to the multiple-choice questions on the answer sheet. No credit will be given for anything written in this exam booklet, but you may use the booklet for notes or scratch work. After you have decided which of the suggested answers is best, completely fill in the corresponding circle on the answer sheet. Give only one answer to each question. If you change an answer, be sure that the previous mark is erased completely. Here is a sample question and answer.

Sample Question	Sample Answer
Chicago is a (A) state (B) city (C) country (D) continent (E) village	Ⓐ ● Ⓒ Ⓓ Ⓔ

Use your time effectively, working as quickly as you can without losing accuracy. Do not spend too much time on any one question. Go on to other questions and come back to the ones you have not answered if you have time. It is not expected that everyone will know the answers to all of the multiple-choice questions.

Your total score on the multiple-choice section is based only on the number of questions answered correctly. Points are not deducted for incorrect answers or unanswered questions.

Form I
Form Code 4MBP4-S

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MACROECONOMICS

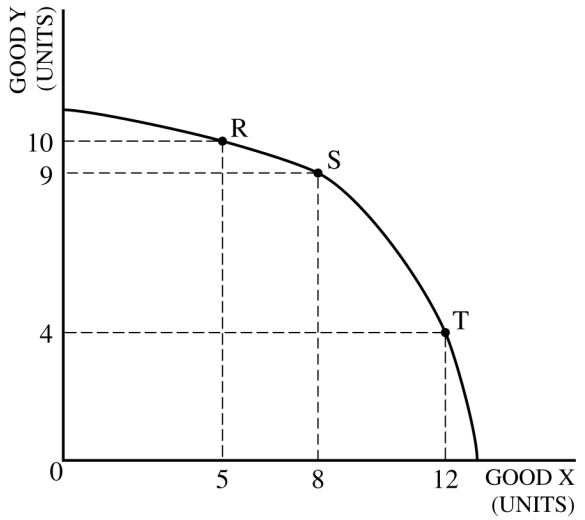
Section I

Time—70 minutes

60 Questions

Directions: Each of the questions or incomplete statements below is followed by five suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

Question 1 is based on the diagram below, which shows the production possibilities curve for an economy that produces only two goods: X and Y.



1. The opportunity cost of moving production from point R to point T is

   (A) one unit of Good Y
   (B) five units of Good Y
   (C) six units of Good Y
   (D) three units of Good X
   (E) seven units of Good X

2. If a country's production possibilities curve is shifting outward, which of the following must be true?

   (A) There is cyclical unemployment.
   (B) The price level is increasing.
   (C) The aggregate demand curve is shifting to the right.
   (D) The long-run Phillips curve is shifting to the right.
   (E) The long-run aggregate supply curve is shifting to the right.

3. In a closed economy with lump sum taxes, if the marginal propensity to consume increased from 0.5 to 0.75, the simple spending multiplier and the marginal propensity to save (MPS) would change in which of the following ways?

	Multiplier	MPS
(A)	Increase	Increase
(B)	Increase	Decrease
(C)	No change	Decrease
(D)	Decrease	Increase
(E)	Decrease	Decrease

Year	Nominal GDP	Price Index
2012	$100,000	200
2013	$110,000	220

4. The table above shows the nominal gross domestic product (GDP) and the price index for an economy during the period 2012 and 2013. Which of the following would have occurred from 2012 to 2013 ?

   (A) Real GDP decreased by 20%.
   (B) Real GDP decreased by 10%.
   (C) Real GDP increased by 20%.
   (D) Real GDP increased by 10%.
   (E) Real GDP did not change.

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5. The equilibrium real interest rate in Britain increases to 8 percent while the equilibrium real interest rate in Australia remains at 4 percent. As a result, financial capital will flow from

(A) Britain to Australia, increasing the interest rate in Britain and decreasing the interest rate in Australia
(B) Britain to Australia, decreasing the interest rate in Britain and increasing the interest rate in Australia
(C) Australia to Britain, increasing the interest rate in Britain and decreasing the interest rate in Australia
(D) Australia to Britain, decreasing the interest rate in Britain and decreasing the interest rate in Australia
(E) Australia to Britain, decreasing the interest rate in Britain and increasing the interest rate in Australia

6. When a central bank conducts open-market bond sales, the money supply, interest rate, and aggregate demand will change in which of the following ways in the short run?

	Money Supply	Interest Rate	Aggregate Demand
(A)	Decrease	Increase	Decrease
(B)	Decrease	Decrease	Increase
(C)	Decrease	Increase	Increase
(D)	Increase	Decrease	Increase
(E)	Increase	Decrease	Decrease

7. If the central bank holds interest rates constant, an autonomous decrease of $10 million in investment spending will most likely result in

(A) a decrease of exactly $10 million in gross domestic product
(B) a decrease of more than $10 million in gross domestic product
(C) an increase of $10 million in taxes to offset the decrease in investment
(D) an increase of $10 million in aggregate supply to offset the decrease in investment
(E) an increase in the cost of loans for investment purposes

8. The circular-flow model indicates that final goods are produced by

(A) firms and sold in the factor markets
(B) firms and sold in the product markets
(C) firms and sold in the resource markets
(D) households and sold in the factor markets
(E) households and sold in the product markets

9. If the interest rate on short-term government bonds declined as a result of open market operations by a central bank, the central bank must have

(A) purchased government bonds
(B) sold government bonds to commercial banks
(C) decreased the amount of currency in circulation
(D) increased the supply of bonds
(E) increased the discount rate on loans to commercial banks

10. Which of the following is most likely to promote long-run economic growth?

(A) An increase in government transfer payments
(B) An increase in income taxes for middle-class households
(C) An increase in tax credits for business spending on research and development
(D) An increase in unemployment compensation
(E) An increase in financial capital outflow

11. To reduce inflation, the central bank would be most likely to

(A) decrease the reserve requirement
(B) decrease the income tax rates
(C) increase the supply of money
(D) buy government securities
(E) sell government securities

12. The implementation of an expansionary monetary policy by the Canadian central bank will result in which of the following changes in the short run?

	Interest Rate	Canadian Dollar
(A)	Increase	Appreciate
(B)	Increase	Depreciate
(C)	Decrease	Depreciate
(D)	No change	Depreciate
(E)	Decrease	Appreciate

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13. The table below shows a country's macroeconomic data in 2013.

Consumption spending	$175 billion
Individual income taxes	$32 billion
Private investment spending	$30 billion
Corporate taxes	$25 billion
Exports	$75 billion
Government purchases	$40 billion
Imports	$100 billion

The country's gross domestic product is

(A) $220 billion
(B) $282 billion
(C) $304 billion
(D) $309 billion
(E) $347 billion

14. Which of the following is a monetary policy that can be used to counteract a recession?

(A) Buying bonds in the open market
(B) Increasing the discount rate
(C) Increasing the required reserve ratio
(D) Lowering tax rates
(E) Increasing government spending

15. If the required reserve ratio is 0.2, a $1 billion increase in bank reserves can lead to an increase in M1 of at most

(A) $6 billion
(B) $5 billion
(C) $1 billion
(D) $0.8 billion
(E) $0.2 billion

Assets			Liabilities	
Reserves		$100,000	Checkable Deposits	$400,000
Required	$80,000			
Excess	$20,000			
Loans		$300,000		
Total		$400,000	Total	$400,000

16. The table above shows the current entries in the T-account of XYZ Bank. Kim purchases a bond issued by the Federal Reserve Bank for $50,000 and pays for the bond by drawing on her company's account at XYZ Bank. What is the effect of Kim's purchase of the bond on the required and excess reserves of XYZ Bank and the total money supply?

	Required	Excess	Money Supply
(A)	Increase	Decrease	Decrease
(B)	Decrease	Decrease	Decrease
(C)	No Change	Decrease	No Change
(D)	Increase	No change	Increase
(E)	Decrease	Increase	Increase

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17. When an economy producing two goods is operating efficiently and at full employment, increasing the production of one good will result in

(A) an inward shift of the production possibilities curve
(B) an outward shift of the production possibilities curve
(C) a decrease in the amount of the other good that can be produced
(D) an increase in the costs of both goods
(E) an increase in the amount of resources available

18. Which of the following describes a major difference between stocks and bonds?

(A) Stocks represent ownership in a corporation, and bonds represent a loan to a corporation.
(B) Bonds represent ownership in a corporation, and stocks represent a loan to a corporation.
(C) Stocks are counted in gross domestic product, and bonds are not counted.
(D) Bonds are counted in gross domestic product, and stocks are not counted.
(E) Bonds pay dividends, and stocks earn interest.

19. When Country X has a trade deficit, which of the following is necessarily true?

(A) Country X has a higher savings rate than most other countries.
(B) The value of Country X's imports exceeds the value of its exports.
(C) Country X has a lower gross domestic product than most other countries.
(D) There is a net financial capital outflow from Country X.
(E) Country X has been increasing tariffs on key goods and services, such as automobiles and steel.

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20. Brazil and Peru produce both coffee and wheat using labor as the only input. The table below shows the labor hours required to produce a unit of coffee and a unit of wheat in each country.

	Labor Hours per Unit of Coffee	Labor Hours per Unit of Wheat
Brazil	12	4
Peru	6	4

Based on the information in the table above, which of the following is true?

(A) Peru has a comparative advantage in producing wheat.
(B) Peru has an absolute advantage in producing wheat.
(C) Brazil has an absolute advantage in producing wheat.
(D) Peru has a comparative advantage in producing coffee.
(E) Brazil has a comparative advantage in producing coffee.

21. Gross private domestic investment includes which of the following?

(A) Changes in business inventories
(B) National defense spending
(C) New durable goods purchased by households
(D) Transfer payments
(E) Depreciation of government goods

22. Which of the following will lead to an increase in United States net exports?

(A) An increase in United States real gross domestic product
(B) Appreciation of the United States dollar on the foreign exchange market
(C) Depreciation of the United States dollar on the foreign exchange market
(D) An increase in government expenditures in the United States
(E) A decrease in income tax rates in the United States

23. An increase in which of the following is most likely to increase employment and promote long-run economic growth?

(A) Government spending on education
(B) Transfer payments
(C) Reserve requirements
(D) Income tax rates
(E) Unemployment compensation

24. Which of the following is implied by a long-run Phillips curve?

(A) Wages are sticky.
(B) There is no trade-off between unemployment and inflation.
(C) The long-run aggregate supply curve is upward-sloping.
(D) Expected inflation exceeds actual inflation.
(E) The unemployment rate falls with higher inflation.

25. Technological progress promotes long-run economic growth primarily by

(A) increasing the supply of financial capital
(B) increasing labor productivity
(C) decreasing national savings
(D) decreasing budget deficit
(E) decreasing nominal wages

26. Which of the following changes would most likely cause an increase in interest rates in the short run?

(A) A decrease in reserve requirements
(B) An increase in trade deficits
(C) An open market purchase of government bonds by the Federal Reserve
(D) An increase in government spending financed by borrowing
(E) An increase in the price of bonds

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27. The short-run Phillips curve shows

(A) a trade-off between the natural rate of unemployment and the rate of inflation
(B) a trade-off between the unemployment rate and rate of inflation
(C) a relationship between increases in the money supply and the rate of inflation
(D) that at the natural rate of unemployment there is no trade-off between the rate of inflation and the unemployment rate
(E) how expectations change as the rate of inflation increases and the unemployment rate decreases

28. In an economy the marginal propensity to consume is 0.90, and gross domestic product (GDP) is $100 billion. If gross private domestic investment declines by $2 billion, then GDP will

(A) decrease by a maximum of $1.8 billion
(B) decrease by a maximum of $2 billion
(C) decrease by a maximum of $20 billion
(D) increase by a maximum of $1.8 billion
(E) increase by a maximum of $20 billion

29. Melanie quits her job to look for a better one that has opportunity for advancement. Melanie will be classified in which category of unemployment?

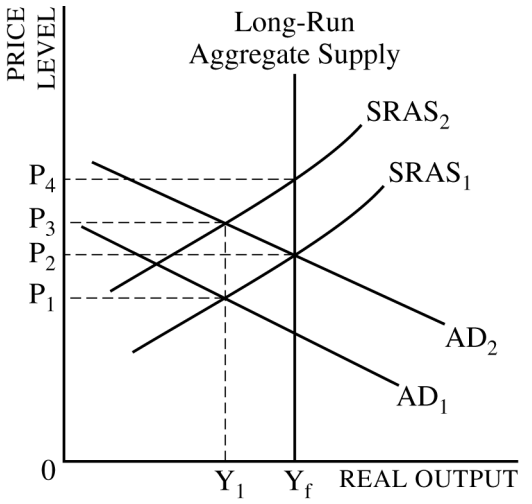
(A) Frictional
(B) Hidden
(C) Seasonal
(D) Structural
(E) Cyclical

30. An increase in money demand will cause which of the following?

(A) A decrease in the nominal interest rate
(B) A decrease in bond prices
(C) A decrease in the money supply
(D) An increase in the price level
(E) An increase in the natural rate of unemployment

31. Which of the following is a best example of an investment in human capital?

(A) Decreasing the wage rate
(B) Training workers
(C) Hiring new workers
(D) Buying new equipment
(E) Updating technology



32. The graph above shows aggregate demand (AD), short-run aggregate supply (SRAS), and long-run aggregate supply curves for an economy. Based on the graph, cost-push inflation is caused by a movement from

(A) $SRAS_1$ to $SRAS_2$
(B) $SRAS_2$ to $SRAS_1$
(C) AD_1 to AD_2
(D) AD_2 to AD_1
(E) Y_1 to Y_f

33. Assume that the reserve requirement is 10 percent. Marwa deposits $1 million in cash into her checking account at First Bank. The deposit will initially increase excess reserves at First Bank by

(A) $100,000
(B) $900,000
(C) $1 million
(D) $9 million
(E) $10 million

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34. Assume a country's government has a balanced budget. If the economy goes into a recession, what will happen to the government's budget in the short run?

(A) It will be in surplus, because the government will increase its spending.
(B) It will be in surplus, because there will be an automatic decrease in unemployment payments.
(C) It will be in deficit, because the government will raise the tax rates.
(D) It will be in deficit, because the price level will decrease.
(E) It will be in deficit, because there will be an automatic decrease in tax receipts.

35. Following a decrease in aggregate demand, an increase in unemployment will result if

(A) prices increase and real wages decrease
(B) the aggregate supply curve is vertical
(C) prices and wages quickly adjust to equilibrium levels
(D) short-run aggregate supply also increases
(E) prices and nominal wages are slow to adjust in the short run

36. In the long run, a fully anticipated increase in the inflation rate will

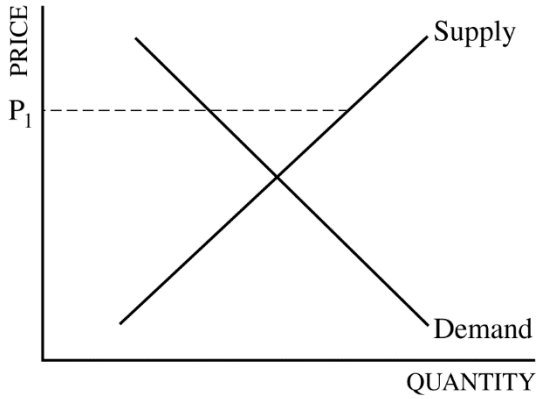
(A) increase real national output
(B) increase the nominal interest rate
(C) decrease real national output
(D) decrease the real interest rate
(E) shift the long-run Phillips curve to the right

37. The exchange rate is 1.2 euros per United States dollar. If a restaurant meal costs 30 euros in Paris, France, what is its dollar cost to a United States tourist?

(A) $2.50
(B) $3.60
(C) $25
(D) $30
(E) $36

38. If the central bank buys government bonds from individuals on the open market and banks do not loan out any excess reserves created by the open market purchase, which of the following will happen?

(A) The money supply will increase.
(B) The money supply will remain unchanged.
(C) Loans to the private sector will increase.
(D) Demand deposits will decrease.
(E) The level of actual reserves will decrease.



39. The graph above shows the demand for and supply of a good. If the market price is P_1, then

(A) there is a shortage, and the price will rise
(B) there is a shortage, and the price will fall
(C) there is a surplus, and the price will rise
(D) there is a surplus, and the price will fall
(E) demand will decrease and supply will increase

40. Assume that the government finances its spending by borrowing from the public. If the government increases deficit spending, the price of previously issued bonds and the real interest rate will change in which of the following ways?

	Price of Bonds	Real Interest Rate
(A)	Decrease	Decrease
(B)	Decrease	Increase
(C)	Increase	Decrease
(D)	Increase	No change
(E)	Increase	Increase

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41. Assume an economy is in a long-run equilibrium. Following a decrease in aggregate demand, which of the following is true in the short run?

(A) The unemployment rate is equal to the natural rate of unemployment, and the rate of inflation is rising.
(B) The unemployment rate is greater than the natural rate of unemployment, and the rate of inflation is rising.
(C) The unemployment rate is greater than the natural rate of unemployment, and the rate of inflation is declining.
(D) The unemployment rate is lower than the natural rate of unemployment, and the rate of inflation is declining.
(E) The unemployment rate is lower than the natural rate of unemployment, and the unemployment rate is rising.

42. An increase in government deficit spending can crowd out private investment by

(A) decreasing the supply of money
(B) increasing the supply of money
(C) decreasing the real interest rate
(D) increasing the real interest rate
(E) decreasing the price level

43. An increase in consumer confidence will result in which of the following in the short run?

(A) A rightward shift of the long-run aggregate supply curve
(B) A rightward shift of the short-run aggregate supply curve
(C) A leftward shift of the short-run aggregate supply curve
(D) A rightward shift of the aggregate demand curve
(E) A leftward shift of the aggregate demand curve

44. Given a constant velocity of money, in the short run a 5 percent increase in money supply will translate to a 5 percent increase in

(A) government budget deficit
(B) real gross domestic product
(C) nominal gross domestic product
(D) real interest rates
(E) nominal interest rates

45. Assume a country has an open economy and a flexible exchange rate system. An increase in the country's government budget deficit would most likely cause

(A) a decrease in real interest rate and a decrease in net exports
(B) a decrease in real interest rate and a decrease in domestic investment in plant and equipment
(C) no change in real interest rate and a decrease in net exports
(D) an increase in real interest rate and an increase in domestic investment in plant and equipment
(E) an increase in real interest rate and a decrease in net exports

46. Which of the following mixes of fiscal and monetary policy would reduce inflation?

	Fiscal Policy	Monetary Policy
(A)	Decrease spending	Buy bonds
(B)	Decrease taxes	Buy bonds
(C)	Increase taxes	Sell bonds
(D)	Increase spending	Buy bonds
(E)	Increase spending	Sell bonds

47. The natural rate of unemployment is the unemployment rate when

(A) all workers seeking employment have taken jobs
(B) all the unemployed are only structurally unemployed
(C) there is no frictional unemployment
(D) there is no cyclical unemployment
(E) there are no new unemployed individuals

48. Which of the following accurately describes the federal funds rate?

(A) The interest rate that banks charge state governments
(B) The interest rate that banks charge other banks for overnight loans
(C) The interest rate that banks pay on long-term savings
(D) The interest rate on personal loans
(E) The interest rate on government bonds

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49. In 2007, the nominal gross domestic product (GDP) was $50 billion and the GDP deflator was 200. Thus real GDP was

(A) $150 billion
(B) $100 billion
(C) $50 billion
(D) $25 billion
(E) $4 billion

50. Supply-side economists are most likely to favor which of the following short-run policies?

(A) Increasing government spending on social welfare
(B) Increasing government spending to help promote the country's business abroad
(C) Cutting marginal tax rates to promote savings, investment, and work
(D) Financing government spending on infrastructure by increasing sales tax rather than increasing income tax
(E) Increasing corporate profit tax rates

51. Nations A and B produce only chairs and bicycles. If each laborer in Nation A can produce twice as many chairs as each laborer in Nation B, then which of the following is necessarily true?

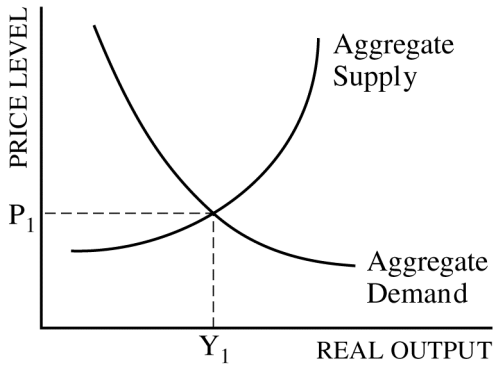
(A) Nation A has a comparative advantage in chairs.
(B) Nation A has a comparative advantage in bicycles.
(C) Nation A has an absolute advantage in chairs.
(D) Nation B has an absolute advantage in bicycles.
(E) Nation A should specialize in producing chairs.

52. If the government implements an expansionary fiscal policy, what action can the central bank take to maintain a stable interest rate?

(A) Increase the required reserve ratio
(B) Increase personal income tax rates
(C) Decrease personal income tax rates
(D) Sell government bonds
(E) Buy government bonds

53. Which of the following is true about the expected real interest rate?

(A) It is equal to the nominal interest rate plus the expected inflation rate.
(B) It is equal to the ratio of the nominal interest rate to the inflation rate.
(C) It increases as the price level increases.
(D) It is always positive.
(E) It is negative if the expected inflation rate exceeds the nominal interest rate.



54. The aggregate supply and aggregate demand graph above shows the current macroeconomic equilibrium of an economy. How will the price level and real output change if there is a sharp increase in productivity and a simultaneous increase in government purchases?

	Price Level	Real Output
(A)	Increase	Increase
(B)	Increase	Decrease
(C)	Indeterminate	Increase
(D)	Indeterminate	Decrease
(E)	Indeterminate	Indeterminate

55. If country X imposes a tariff on its imports, how will the supply of its currency and its exchange rate be affected in foreign exchange markets?

	Supply	Currency
(A)	Increase	Depreciate
(B)	Increase	Appreciate
(C)	No change	Appreciate
(D)	Decrease	Depreciate
(E)	Decrease	Appreciate

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56. In the short run, cost-push inflation can be caused by

(A) printing too much money
(B) increased government spending
(C) negative supply shocks
(D) decreased tariffs
(E) decreased taxes

57. Which of the following necessarily occurs during an economic recession?

(A) Cost-push inflation decreases.
(B) Real gross domestic product decreases.
(C) Cyclical unemployment decreases.
(D) Demand-pull inflation increases.
(E) Nominal wages increase.

58. Which of the following will cause an increase in real output in the short run?

(A) A decrease in productivity
(B) A decrease in wages
(C) A decrease in government expenditure
(D) An increase in reserve requirements
(E) An increase in the discount rate

59. A lender will realize unexpected benefit when the

(A) actual inflation rate is higher than the anticipated inflation rate
(B) actual inflation rate is lower than the anticipated inflation rate
(C) rate of interest is greater than the actual rate of inflation
(D) rate of interest is less than the actual rate of inflation
(E) rate of interest equals the actual rate of inflation

60. An increase in the money supply will affect the price level and real gross domestic product (GDP) in which of the following ways in the long run?

	Price Level	Real GDP
(A)	Decrease	No change
(B)	Increase	Decrease
(C)	Increase	No change
(D)	Decrease	Increase
(E)	No change	No change

END OF SECTION I

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION.

DO NOT GO ON TO SECTION II UNTIL YOU ARE TOLD TO DO SO.

MAKE SURE YOU HAVE DONE THE FOLLOWING.

- **PLACED YOUR AP NUMBER LABEL ON YOUR ANSWER SHEET**
- **WRITTEN AND GRIDDED YOUR AP NUMBER CORRECTLY ON YOUR ANSWER SHEET**
- **TAKEN THE AP EXAM LABEL FROM THE FRONT OF THIS BOOKLET AND PLACED IT ON YOUR ANSWER SHEET**

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Section II: Free-Response Questions

This is the free-response section of the 2016 AP exam.
It includes cover material and other administrative instructions
to help familiarize students with the mechanics of the exam.
(Note that future exams may differ in look from the following content.)

AP® Macroeconomics Exam

SECTION II: Free Response

2016

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

At a Glance

Total Time
1 hour
Number of Questions
3
Percent of Total Score
33.33%
Writing Instrument
Pen with black or dark blue ink
Electronic Device
None allowed

Reading Period

Time
10 minutes. Use this time to read the questions and plan your answers. You may begin writing your responses before the reading period is over.

Writing Period

Time
50 minutes

Question 1
Suggested Time
25 minutes
Percent of Section II Score
50%

Question 2
Suggested Time
12.5 minutes
Percent of Section II Score
25%

Question 3
Suggested Time
12.5 minutes
Percent of Section II Score
25%

IMPORTANT Identification Information

PLEASE PRINT WITH PEN:

1. First two letters of your last name ☐☐
   First letter of your first name ☐
2. Date of birth
   ☐☐ Month ☐☐ Day ☐☐☐☐ Year
3. Six-digit school code
   ☐☐☐☐☐☐
4. Unless I check the box below, I grant the College Board the unlimited right to use, reproduce, and publish my free-response materials, both written and oral, for educational research and instructional purposes. My name and the name of my school will not be used in any way in connection with my free-response materials. I understand that I am free to mark "No" with no effect on my score or its reporting.
   No, I do not grant the College Board these rights. ☐

Instructions

The questions for Section II are printed in this booklet. You may use page 3 and the pages the questions are printed on to organize your answers and for scratch work, but you must write your answers on the lined pages provided for each question.

The proctor will announce the beginning and end of the reading period. You are advised to spend the 10-minute period reading all the questions and planning your answers. You may begin writing your responses before the reading period is over.

Write clearly and legibly. Do not skip lines. Cross out any errors you make; crossed-out work will not be scored.

Manage your time carefully. You may proceed freely from one question to the next. You may review your responses if you finish before the end of the exam is announced.

Form I
Form Code 4MBP4-S

MACROECONOMICS

Section II

Total Time—60 minutes

Reading Period—10 minutes

Writing Period—50 minutes

Directions: You are advised to spend the first 10 minutes reading all of the questions and planning your answers. You will then have 50 minutes to answer all three of the following questions. You may begin writing your responses before the reading period is over. It is suggested that you spend approximately half your time on the first question and divide the remaining time equally between the next two questions. Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. Use a pen with black or dark blue ink.

Question 1 begins on page 4.

Question 2 begins on page 10.

Question 3 begins on page 14.

THIS PAGE MAY BE USED FOR TAKING NOTES AND PLANNING YOUR ANSWERS.

NOTES WRITTEN ON THIS PAGE WILL NOT BE SCORED.

WRITE ALL YOUR RESPONSES ON THE LINED PAGES.

1. The economy of Zarland is operating below the full-employment level of output with a balanced budget.

   (a) Draw a correctly labeled graph of short-run aggregate supply, long-run aggregate supply, and aggregate demand, and show each of the following.

      (i) The country's current equilibrium output and price level, labeled Y_1 and PL_1, respectively

      (ii) The full-employment output, labeled Y_f

   (b) If Zarland increases government expenditures and taxes by equal amounts, can aggregate demand increase? Explain.

   (c) If Zarland decides to pursue an expansionary monetary policy, what open-market operation should the central bank undertake?

   (d) Using a correctly labeled graph of the money market, show the short-run effect of the open-market operation you identified in part (c) on the interest rate.

   (e) As a result of the interest rate change associated with the expansionary monetary policy, will the supply of Zarland's currency in the foreign exchange market increase, decrease, or remain the same in the short run? Explain.

   (f) Based on your answer in part (e), will the value of Zarland's currency in the foreign exchange market increase, decrease, or remain the same in the short run?

   (g) Following the change in the value of Zarland's currency that you identified in part (f), will Zarland's exports increase, decrease, or remain the same in the short run? Explain.

THIS PAGE MAY BE USED FOR TAKING NOTES AND PLANNING YOUR ANSWERS.

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ANSWER PAGE FOR QUESTION 1

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Question 1 is reprinted for your convenience.

1. The economy of Zarland is operating below the full-employment level of output with a balanced budget.

   (a) Draw a correctly labeled graph of short-run aggregate supply, long-run aggregate supply, and aggregate demand, and show each of the following.

      (i) The country's current equilibrium output and price level, labeled Y_1 and PL_1, respectively

      (ii) The full-employment output, labeled Y_f

   (b) If Zarland increases government expenditures and taxes by equal amounts, can aggregate demand increase? Explain.

   (c) If Zarland decides to pursue an expansionary monetary policy, what open-market operation should the central bank undertake?

   (d) Using a correctly labeled graph of the money market, show the short-run effect of the open-market operation you identified in part (c) on the interest rate.

   (e) As a result of the interest rate change associated with the expansionary monetary policy, will the supply of Zarland's currency in the foreign exchange market increase, decrease, or remain the same in the short run? Explain.

   (f) Based on your answer in part (e), will the value of Zarland's currency in the foreign exchange market increase, decrease, or remain the same in the short run?

   (g) Following the change in the value of Zarland's currency that you identified in part (f), will Zarland's exports increase, decrease, or remain the same in the short run? Explain.

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ADDITIONAL PAGE FOR ANSWERING QUESTION 1

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ADDITIONAL PAGE FOR ANSWERING QUESTION 1

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ADDITIONAL PAGE FOR ANSWERING QUESTION 1

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2. Assume that the current unemployment rate in Country A is lower than the natural rate of unemployment.

   (a) Draw a single correctly labeled graph with both the long-run Phillips curve and the short-run Phillips curve. Label the current short-run equilibrium point Z.

   (b) Identify a specific fiscal policy action that would bring the economy to full employment.

   (c) Draw a correctly labeled graph of the loanable funds market, and show the effect of the fiscal policy from part (b) on the real interest rate in the short run.

   (d) Now assume instead that there is no fiscal policy action. Will the short-run Phillips curve shift to the right, shift to the left, or remain the same over time? Explain.

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ANSWER PAGE FOR QUESTION 2

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ADDITIONAL PAGE FOR ANSWERING QUESTION 2

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ADDITIONAL PAGE FOR ANSWERING QUESTION 2

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3. The Bank of Uchenna has the following balance sheet.

Assets		Liabilities and Equity	
Reserves	$30,000	Demand deposits	$100,000
Loans	$70,000		
Property	$2,000	Equity (net worth)	$2,000

Assume that the reserve requirement ratio is 20 percent.

(a) Calculate the dollar value of the reserves that the Bank of Uchenna is required to hold.

(b) Given the current reserves, calculate the maximum value of additional loans that the Bank of Uchenna can make.

(c) Assume that Elike raises $5,000 in cash from a yard sale and deposits the cash in his checking account at the Bank of Uchenna. By how much does the money supply immediately change as a result of Elike's deposit?

(d) Calculate the maximum change in demand deposits in the banking system as a whole resulting from Elike's deposit.

(e) If the Bank of Uchenna is not meeting its reserve requirement, what action can it take to meet the reserve requirement without calling in loans or selling property?

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ANSWER PAGE FOR QUESTION 3

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ADDITIONAL PAGE FOR ANSWERING QUESTION 3

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ADDITIONAL PAGE FOR ANSWERING QUESTION 3

GO ON TO THE NEXT PAGE.

STOP

END OF EXAM

THE FOLLOWING INSTRUCTIONS APPLY TO THE COVERS OF THE SECTION II BOOKLET.

- **MAKE SURE YOU HAVE COMPLETED THE IDENTIFICATION INFORMATION AS REQUESTED ON THE FRONT AND BACK COVERS OF THE SECTION II BOOKLET.**
- **CHECK TO SEE THAT YOUR AP NUMBER LABEL APPEARS IN THE BOX ON THE COVER.**
- **MAKE SURE YOU HAVE USED THE SAME SET OF AP NUMBER LABELS ON ALL AP EXAMS YOU HAVE TAKEN THIS YEAR.**

Multiple-Choice Answer Key

The following contains the answers to
the multiple-choice questions in this exam.

Answer Key for AP Macroeconomics Practice Exam, Section I

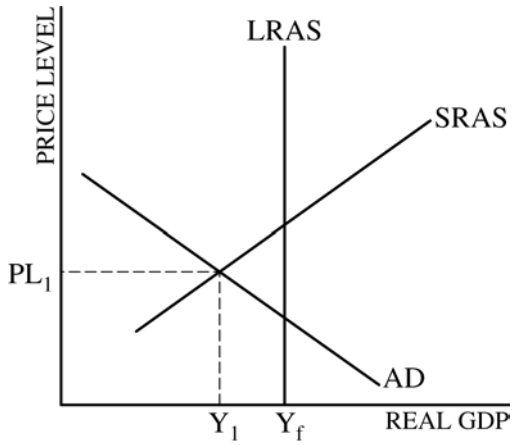
Question 1: C
Question 2: E
Question 3: B
Question 4: E
Question 5: E
Question 6: A
Question 7: B
Question 8: B
Question 9: A
Question 10: C
Question 11: E
Question 12: C
Question 13: A
Question 14: A
Question 15: B
Question 16: B
Question 17: C
Question 18: A
Question 19: B
Question 20: D
Question 21: A
Question 22: C
Question 23: A
Question 24: B
Question 25: B
Question 26: D
Question 27: B
Question 28: C
Question 29: A
Question 30: B
Question 31: B
Question 32: A
Question 33: B
Question 34: E
Question 35: E
Question 36: B
Question 37: C
Question 38: A
Question 39: D
Question 40: B
Question 41: C
Question 42: D
Question 43: D
Question 44: C
Question 45: E
Question 46: C
Question 47: D
Question 48: B
Question 49: D
Question 50: C
Question 51: C
Question 52: E
Question 53: E
Question 54: C
Question 55: E
Question 56: C
Question 57: B
Question 58: B
Question 59: B
Question 60: C

Free-Response Scoring Guidelines

The following contains the scoring guidelines for the free-response questions in this exam.

Question 1

9 points: 2+1+1+2+1+1+1



(a) 2 points:

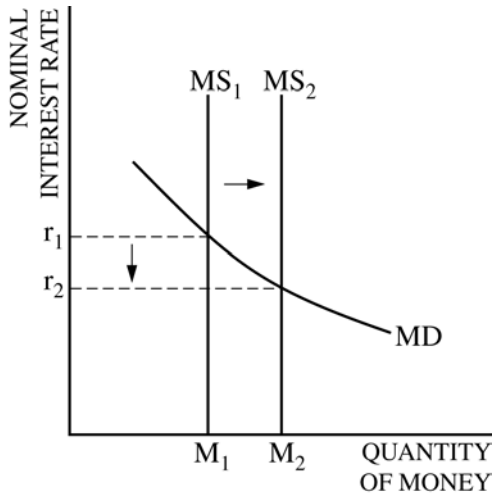
- One point is earned for drawing a correctly labeled graph showing AD, SRAS, Y_1 and PL_1.
- One point is earned for drawing a vertical LRAS at Y_f to the right of Y_1.

(b) 1 point:

- One point is earned for stating that aggregate demand can increase, and for explaining that government spending multiplier (1/mps) is greater than the tax multiplier (mpc/mps), therefore an equal increase in government expenditures and taxes will increase real GDP by the amount of the initial increase in government expenditures.

(c) 1 point:

- One point is earned for stating that the central bank should buy bonds.



(d) 2 points:

- One point is earned for drawing a correctly labeled money market graph.
- One point is earned for shifting the money supply curve to the right and showing a decrease in the nominal interest rate.

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Question 1 (continued)

(e) 1 point:

- One point is earned for stating that the supply of Zarland's currency will increase and for explaining that financial investors seeking higher returns for their financial investments will sell Zarland's financial assets (capital outflow) to buy foreign assets.

(f) 1 point

- One point is earned for stating that the value of Zarland's currency will decrease (or Zarland's currency will depreciate).

(g) 1 point

- One point is earned for stating that Zarland's exports will increase and for explaining that Zarland's goods will become relatively cheaper than foreign goods.

  OR/ because the depreciation in Zarland's currency means appreciation in foreign currencies which increases the demand for Zarland's goods.

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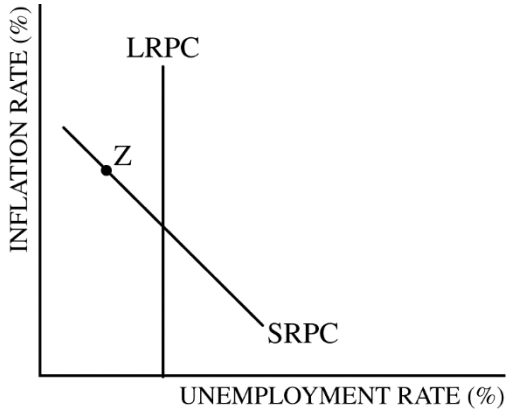
Question 2

6 points: 2+1+2+1

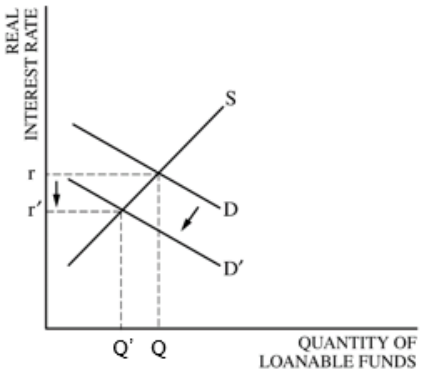


(a) 2 points:

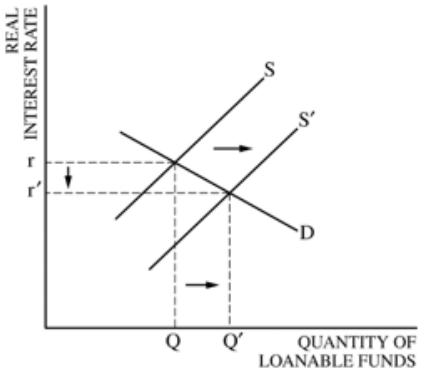
- One point is earned for drawing a correctly labeled graph showing a vertical LRPC and a downward sloping SRPC.
- One point is earned for showing point Z on the SRPC to the left of LRPC.

(b) 1 point:

- One point is earned for stating that a contractionary fiscal policy such as an increase in taxes or a decrease in government expenditure will return the economy to full employment.



OR



(b) 2 points:

- One point is earned for drawing a correctly labeled graph of the loanable funds market.
- One point is earned for shifting the supply of loanable funds curve to the right or the demand for loanable funds to the left and showing a decrease in the real interest rate.

(d) 1 point:

- One point is earned for stating that SRPC will shift to the right (upwards) and for explaining that inflationary expectations increase, or input prices increase, or wages increase.

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Question 3

5 points: 1+1+1+1+1

(a) 1 point:
- One point is earned for calculating the required reserves: $20,000.

(b) 1 point:
- One point is earned for calculating the maximum amount of additional loans that Bank of Uchenna can make as $10,000 (= $30,000 – $20,000).

(c) 1 point:
- One point is earned for stating that there will be no change in the money supply.

(d) 1 point:
- One point is earned for stating that the maximum change in demand deposits in the banking system is $25,000 (= $5,000 x 5).

(e) 1 point:
- One point is earned for stating that Bank of Uchenna can borrow from other banks to meet its reserve requirements or borrow from the central bank.

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Scoring Worksheet

The following provides a scoring worksheet and conversion table used for calculating a composite score of the exam.

2016 AP Macroeconomics Scoring Worksheet

Section I: Multiple Choice

_____	× 1.0000 =	_____
Number Correct (out of 60)		Weighted Section I Score (Do not round)

Section II: Free Response

Question 1	_____ (out of 9)	× 1.6666 =	_____ (Do not round)
Question 2	_____ (out of 6)	× 1.2500 =	_____ (Do not round)
Question 3	_____ (out of 5)	× 1.5000 =	_____ (Do not round)

Sum = _____
Weighted Section II Score (Do not round)

Composite Score

_____	+	_____	=	_____
Weighted Section I Score		Weighted Section II Score		Composite Score (Round to nearest whole number)

AP Score Conversion Chart
Macroeconomics

Composite Score Range	AP Score
70-90	5
57-69	4
47-56	3
35-46	2
0-34	1

Question Descriptors and Performance Data

The following contains tables showing the content assessed,
the correct answer, and how AP students performed on each question.

2016 AP Macroeconomics
Question Descriptors and Performance Data

Multiple-Choice Questions

Question	Topic	Key	% Correct
1	Basic Economic Concepts	C	82
2	Economic Growth And Productivity	E	64
3	National Income And Price Determination	B	74
4	Measurement Of Economic Performance	E	39
5	Open Economy: International Trade And Finance	E	45
6	Inflation\Unemployment And Stabilization Policies	A	58
7	National Income And Price Determination	B	45
8	Measurement Of Economic Performance	B	83
9	Financial Sector	A	55
10	Economic Growth And Productivity	C	55
11	Inflation\Unemployment And Stabilization Policies	E	67
12	Open Economy: International Trade And Finance	C	48
13	Measurement Of Economic Performance	A	61
14	Inflation\Unemployment And Stabilization Policies	A	68
15	Financial Sector	B	59
16	Financial Sector	B	43
17	Basic Economic Concepts	C	82
18	Financial Sector	A	76
19	Open Economy: International Trade And Finance	B	83
20	Basic Economic Concepts	D	55
21	Measurement Of Economic Performance	A	45
22	Open Economy: International Trade And Finance	C	72
23	Economic Growth And Productivity	A	82
24	Inflation\Unemployment And Stabilization Policies	B	46
25	Economic Growth And Productivity	B	86
26	Inflation\Unemployment And Stabilization Policies	D	47
27	Inflation\Unemployment And Stabilization Policies	B	76
28	National Income And Price Determination	C	54
29	Measurement Of Economic Performance	A	77
30	Financial Sector	B	21
31	Economic Growth And Productivity	B	71
32	Inflation\Unemployment And Stabilization Policies	A	63
33	Financial Sector	B	60
34	Inflation\Unemployment And Stabilization Policies	E	37
35	National Income And Price Determination	E	46
36	Inflation\Unemployment And Stabilization Policies	B	38
37	Open Economy: International Trade And Finance	C	65
38	Financial Sector	A	43

2016 AP Macroeconomics

Question Descriptors and Performance Data

Question	Topic	Key	% Correct
39	Basic Economic Concepts	D	73
40	Inflation\Unemployment And Stabilization Policies	B	37
41	Measurement Of Economic Performance	C	59
42	Inflation\Unemployment And Stabilization Policies	D	58
43	National Income And Price Determination	D	83
44	Financial Sector	C	42
45	Open Economy: International Trade And Finance	E	48
46	Inflation\Unemployment And Stabilization Policies	C	62
47	Measurement Of Economic Performance	D	58
48	Financial Sector	B	65
49	Measurement Of Economic Performance	D	51
50	Inflation\Unemployment And Stabilization Policies	C	54
51	Basic Economic Concepts	C	65
52	Inflation\Unemployment And Stabilization Policies	E	32
53	Financial Sector	E	34
54	National Income And Price Determination	C	63
55	Open Economy: International Trade And Finance	E	34
56	Inflation\Unemployment And Stabilization Policies	C	54
57	Basic Economic Concepts	B	77
58	National Income And Price Determination	B	41
59	Measurement Of Economic Performance	B	48
60	National Income And Price Determination	C	44

AP Macroeconomics

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